Phase 1 reprint, set 1: wagon, points and handover

Version 2, checked in FreeCAD. Sliced files for the Bambu Lab P1S are in the same folder. The 7 parts from the first print that you keep are listed on the right.

12 files
to print

27 pieces
on the bed

422 g
of filament

12h 07m
of printing

Filament to have ready (reprint set)

PLA Light Grey	216 g (buy with 20 % margin: 259 g)
PLA Cobalt Blue	154 g (buy with 20 % margin: 185 g)
PLA Orange	46 g (buy with 20 % margin: 55 g)
PETG Orange	7 g (buy with 20 % margin: 8 g)

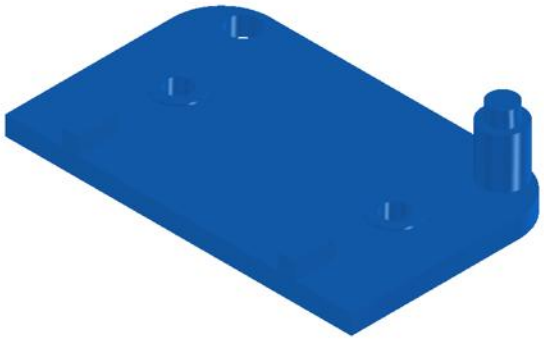
Keep from the first print

fit_coupon_x1: use your printed fit_coupon_x1
wheel_x6: use your printed wheel_x6
wedge_x1: use your printed wedge_x2 (you have 2, need 1)
lead_rail_230_x3: use your printed lead_rail_230_x3
chain_11_links_x3: use your printed chain_11_links_x3
gate_key_x1: use your printed gate_key_x4 (you have 4, need 1)
gate_pin_x1: use your printed gate_pin_x3 (you have 3, need 1)

Print order

1. Wagon parts first: plates, panniers, pins (about 4.7 h). Build one wagon with your existing wheels and roll it on a lead rail.
2. Then the points: turnout plate, tongue, rocker bridge (with supports), rocker, pins, spring, flap, axle; then the handover.
3. Work the points by hand before printing set 2.

Reprint parts 1 to 3 of 12



PRINT Wagon

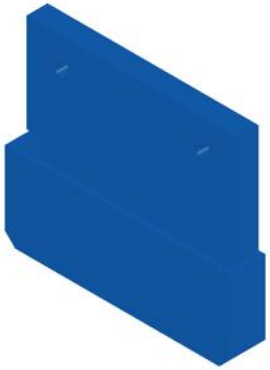
trolley_plate_x6

PLA Cobalt Blue

qty 6 · 49.1 g · 1h 27m · 64.0 × 40.0 × 15.8 mm on the bed · 0.20 mm layers

two identical plates per wagon; prints flat, peg up

Why reprint: pegs reach the other plate and carry shoulders that set the 9.2 mm gap; axle holes print round (the plate lies flat); lips 0.5 mm clear of the web; 3.6 thick with 0.4 bosses to suit 0.2 mm layers



PRINT Wagon

pannier_x6

PLA Cobalt Blue

qty 6 · 105.2 g · 2h 25m · 64.0 × 10.2 × 54.8 mm on the bed · 0.20 mm layers

basket; two identical per wagon, clamped by the axle pins

Why reprint: new: one part for both sides, clamped by the axle pins, floor 12 mm above the strip; arm spacing for the 9.2 mm plate gap



PRINT Wagon

axle_pin_x6

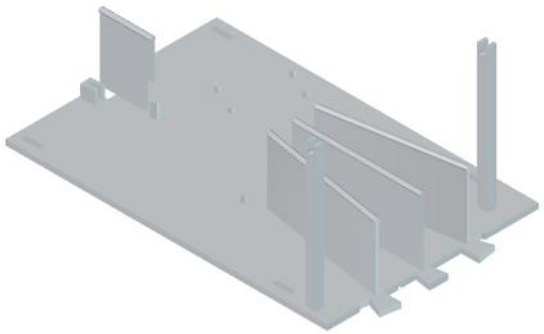
PETG Orange

qty 6 · 3.4 g · 0h 52m · 8.0 × 8.0 × 32.4 mm on the bed · 0.12 mm layers

4 mm pin through pannier, plate, wheel, plate, pannier; snap bump

Why reprint: longer (28.4 mm shank) to pass through pannier, plate, wheel, plate, pannier; snap bump 0.7 mm clear

Reprint parts 4 to 6 of 12



PRINT Points

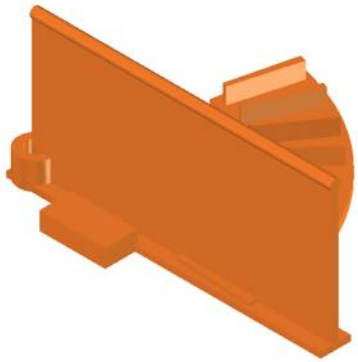
turnout_base_x1

PLA Light Grey

qty 1 · 159.9 g · 3h 38m · 249.0 × 155.0 × 109.0 mm on the bed · 0.20 mm layers

plate, entry and exit rails, pivot, stops, rocker bridges, gantry posts

Why reprint: new layout: lane spacing, pegs for the rocker bridge, gantry posts with 3.6 mm snap mouths, wire groove underneath; the entry rail's end lets the tongue slide under



PRINT Points

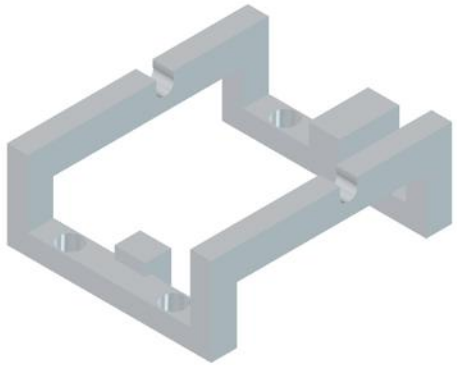
tongue_x1

PLA Orange

qty 1 · 22.5 g · 0h 41m · 106.0 × 88.7 × 49.5 mm on the bed · 0.20 mm layers

switch blade with the ring of four teeth

Why reprint: new: ring of four teeth and the spring tab



PRINT Points

rocker_bridge_x1

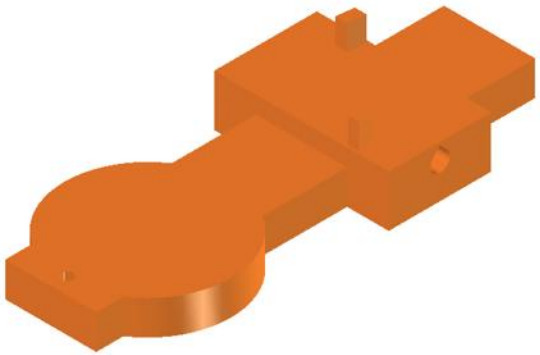
PLA Light Grey

qty 1 · 6.6 g · 0h 25m · 39.4 × 53.0 × 17.0 mm on the bed · 0.20 mm layers

two beams over the tooth ring; carries the rocker pin and the tilt stops; sits on four pegs

Why reprint: new: carries the rocker over the tooth ring; separate so the tongue can be fitted first; beams 2.5 mm over the teeth, sliced with supports

Reprint parts 7 to 9 of 12



PRINT Points

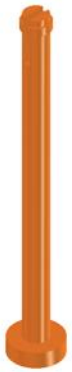
rocker_x1

PLA Orange

qty 1 · 8.3 g · 0h 28m · 30.6 × 77.0 × 13.4 mm on the bed · 0.12 mm layers

escapement rocker; prints upside down

Why reprint: new: escapement rocker with a 26 mm coin cup



PRINT Points

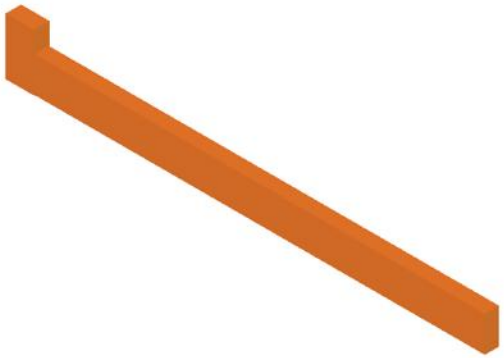
rocker_pin_x1

PETG Orange

qty 1 · 0.5 g · 0h 14m · 6.5 × 6.5 × 44.0 mm on the bed · 0.12 mm layers

3 mm rocker axle with snap bump

Why reprint: new



PRINT Points

bias_spring_x1

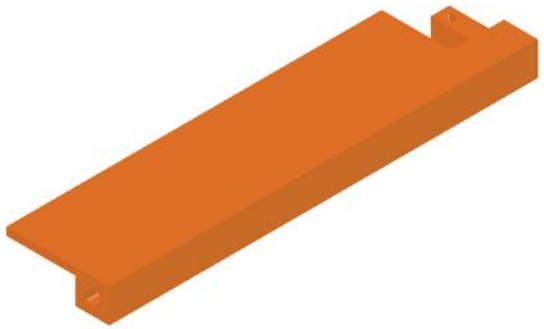
PETG Orange

qty 1 · 1.0 g · 0h 14m · 67.4 × 2.2 × 8.0 mm on the bed · 0.12 mm layers

leaf spring that pushes the tongue toward the next lane

Why reprint: new leaf spring

Reprint parts 10 to 12 of 12



PRINT Points

flap_x1

PLA Orange

qty 1 · 14.8 g · 0h 25m · 34.5 × 114.0 × 9.0 mm on the bed · 0.20 mm layers

trigger flap with crank; prints on its downstream face

Why reprint: new trigger flap with a 16 mm crank



PRINT Points

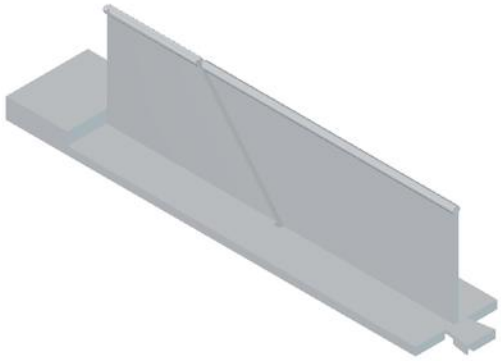
gantry_axle_x1

PETG Orange

qty 1 · 2.1 g · 0h 12m · 4.0 × 134.0 × 3.5 mm on the bed · 0.12 mm layers

4 mm flap axle, prints on its flat

Why reprint: new



PRINT Wire end

handover_x1

PLA Light Grey

qty 1 · 49.1 g · 1h 06m · 212.0 × 40.0 × 53.5 mm on the bed · 0.20 mm layers

wire-to-rail handover; the wire dives through it

Why reprint: the wire now dives through it and runs on under the yard; head height matches the wheel seat